Certicom 131-bit index-calculus program: gates and two falsifiers

Study `research/certicom131_ic_program_20261020/`. **Program text** `docs/ic/CERTICOM131_IC_PROGRAM.md`. **Class:** measurement / accounting. **Date:** 2026-10-20. Public synthetic instances only; the Certicom challenge targets are named to fix parameters and the rho reference, and nothing is run against them. No sub-rho claim, no key-recovery claim, no statement about a deployed curve.

Summary (measured / derived / extrapolated, labelled)

- **Derived (cost model, `gates.md`).** Matched rho: ECC2-131 $2^{64.83}$, ECCp-131 $2^{64.92}$ GAE (negation map). Binary FPPR with $m \le 3$ cannot reach rho at $n = 131$ even with a free decomposition oracle (floor $2^{65.2}$ to $2^{67.2}$). The first usable arity is $m = 4$ with a solving-degree gate $D^* \le 3$ (generous price) or ≤ 1 (plain); $m = 5, 6$ allow D^* up to 4 to 5 only at the generous price and favourable width. PKM tower on ECCp-131: $D^* \le 3$ at $m = 4, 5$. Small-abscissa base: needs $\delta \ge 0.169$ at $m = 4$ against a lattice reach of 0.0125 (13.5x shortfall).
- **Measured, Falsifier A.** Plain-Macaulay resolving degree at $m = 3$ on random binary curves is $D = 4, 5, 6$ for $\ell = 1, 2, 3$ (60 draws, zero spread), equal to the semi-regular reference in every cell, and equal to the Koblitz $b = 1$ value on the same (V, x_R) in 52 of 52 paired draws. First fall degree stays 2 to 3. No scaling deviation from a random b.
- **Measured, Falsifier B.** The combined-system count of structured prime-field bases (Kummer μ_{2^t}, coset, subgroup, interval) is within 3% of the random-oracle expectation and of a random subset: null.
- **Measured, ecbench.** New 30-bit prime-kummer row: IC is 159x matched rho at $w = 0$ (121.6 to 221.4), 175x at $w = 0.1$, 247x at $w = 1$, up from 140.6x at 28 bits. Audit: 160 records, 160 verified, 120 replays reproduced.
- **Extrapolated, not measured.** D at $n = 131$, $m = 3$ between about 6 and 7 (plateau) and 39 (semi-regular); the PKM gap near $\log_2 r = 130$ of order 10^7 if the $r^{1/6}$ law holds.

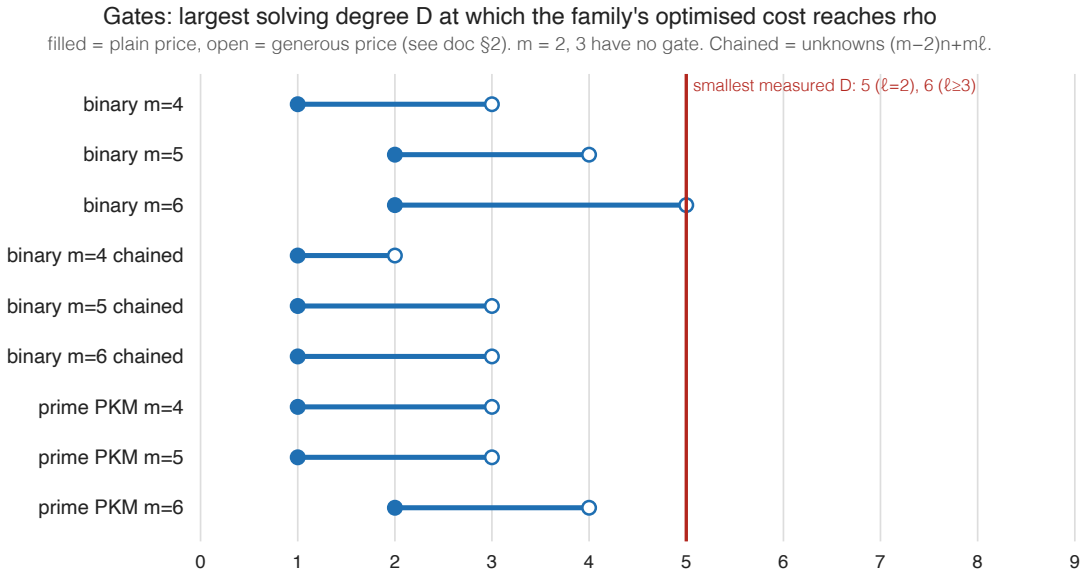


Figure 1: Gates from `certicom131_ic_gates` (`gates.json`): largest solving degree D at which the family's optimised cost reaches rho, at the plain (filled) and generous (open) price points.

Falsifier A: resolving degree on random binary curves

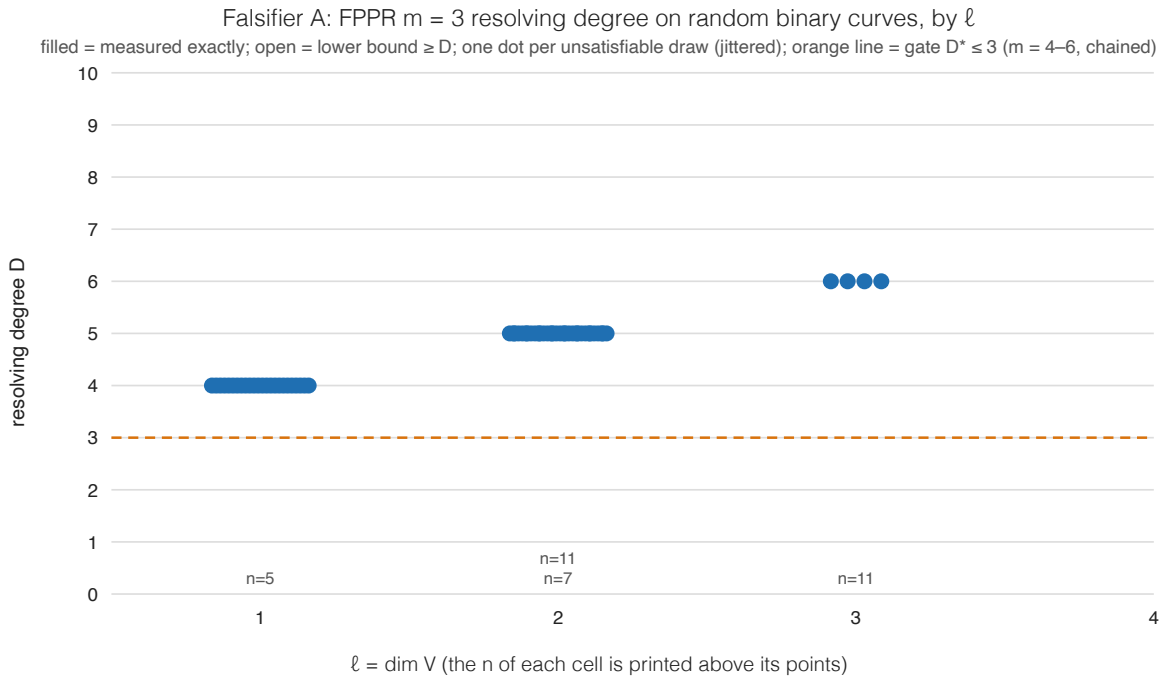


Figure 2: Run `certicom131_fppr_dreg`, seed 20261020, 60 unsatisfiable draws (each verified unsatisfiable by exact count). Records `runs/c*.jsonl`.

n	ℓ	S	unknowns	draws	random-b D	b=1 D (same V, x_R)	FFD	semi-reg.
5	1	2	8	24	4 ×24	4 ×20 (4 sat)	2–3	4
7	2	1	13	24	5 ×24	5 ×23 (1 sat)	2–3	5
11	2	5	17	8	5 ×8	5 ×8	3	5
11	3	2	20	4	6 ×4	6 ×1 (3 unpaired)	3	6

Not run: cells (13,3), (13,4), (17,5), (19,6) were launched and produced no draw in about 3 h of wall time at host load 90 to 170. The $n \leq 63$ cap of the instrument is reachable for the exact count, not for the Macaulay measurement (a draw costs about 10^3 x more from $\ell = 2$ to $\ell = 3$). Wall times are a practicality note only.

Falsifier B: combined-system solution count

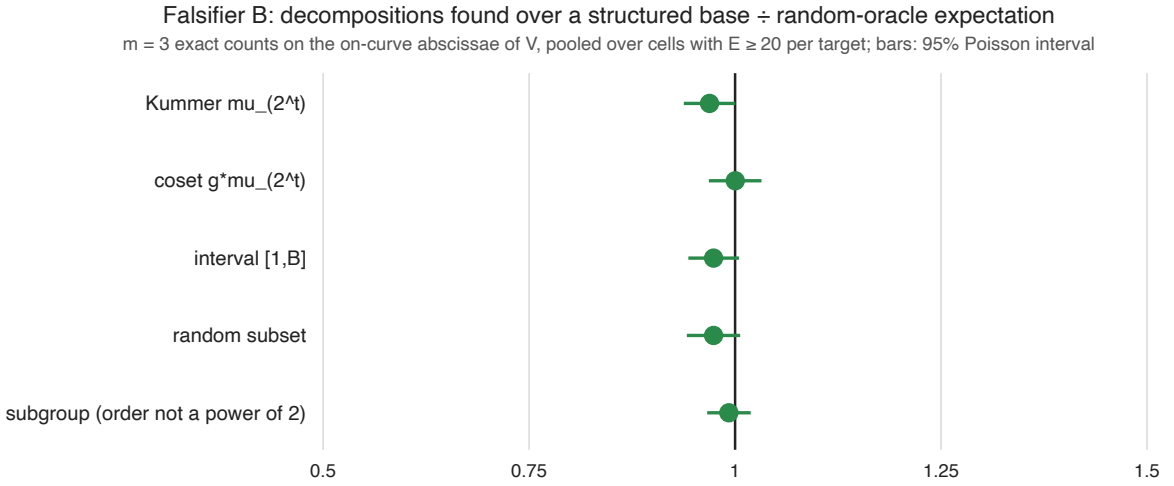


Figure 3: Run `certicom131_prime_combined_count`, seed 20261020, four curves of the PKM ladder (20 to 28 bit), eight on-curve targets, $m = 3$, $B = 2^6$ to 2^{11}; cells with expected count ≥ 20 per target pooled. Records `runs/fb_*.jsonl`.

base	cells	Σ N_3	Σ E_3	N/E	95% interval
coset $g\mu$	8	11,324	11,322.7	1.000	0.968, 1.032

subgroup, non-2-power order	7	16,348	16,470.8	0.993	0.966, 1.019
interval [1,B]	8	11,591	11,901.4	0.974	0.943, 1.005
random subset (control)	7	10,445	10,726.1	0.974	0.941, 1.006
Kummer μ_{2^t}	7	11,107	11,461.6	0.969	0.938, 1.000

The note's dead criterion $N \geq B^{m-1}$ is never met: $N_3 \leq 560$ per target against $B^2 \geq 2.5 \cdot 10^5$.

ecbench: prime-kummer past 28 bits

Session `ECBS1hfb8f08d66f01`, spec `ECS1h5e999ae301b7`, curve `icv1-fp30-t49927-d70f5de2` ($p = 1000574977$, order 1000525051 (prime), $\log_2 r = 29.90$), Kummer $t = 12$ (2,092 columns), 8 targets x 3 rounds, L0 (operation counts only, host loaded). Mean S at w = 0, 0.1, 1 (in that order): rho-neg 0.921 / 1.012 / 1.830; ic-rootfind 146.4 / 177.0 / 452.2; ic-mitm 144.7 / 175.6 / 454.2; ic-abscissa control 146.6 / 179.7 / 477.5. Tower base equals plain base within 1.2%. The 32-bit row was not run (curve generator's exhaustive point count unfinished).

Limits

Toy sizes ($n \leq 11$, $\ell \leq 3$; $p \leq 2^{30}$); extrapolation to 131 bits is a factor 6 to 10 outside the measured ℓ range and is labelled as such. The model's constants are assumptions with two bracketing price points. Host: shared macOS machine at load 90 to 170, no isolation (L0); wall time is not a metric here.